EXP No: 7 FREQUENCY RESPONSE OF SERIES AND PARALLEL RESONANCE CIRCUITS

DATE 11/8/26

Pg. No

AIM:

To study the behaviours of series and parallel RLC circuits at resonance and to determine the resonant frequency, bandwidth and Q of the given RLC circuit using multisim simulator.

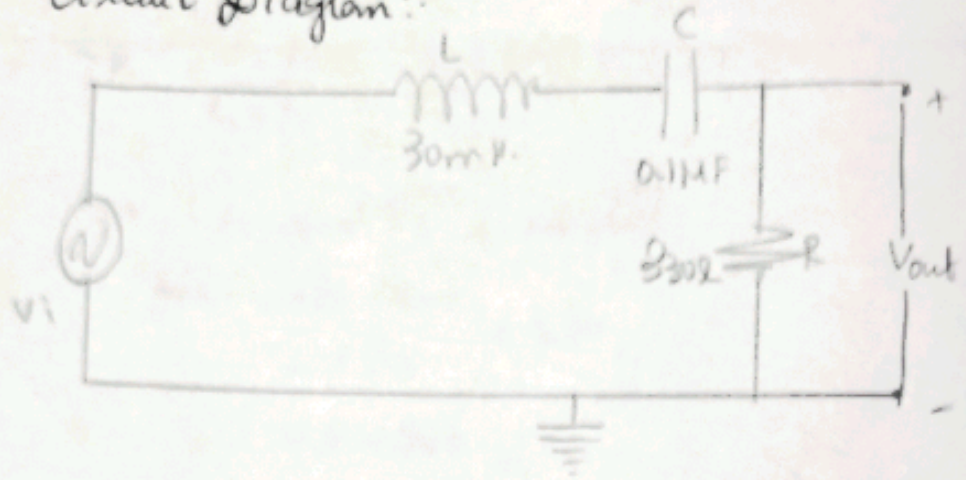
APPARATUS REQUIRED:

S.No	Components	Specification	Quantity
1	CRO	10 MHz	1
2	DIB	–	1
3	Capacitors	0.1 μf	1
4	Resistors	330 Ω	1
5	Bread Board	–	–
6	connecting wires	–	–
7	PC with multisim simulator	–	–

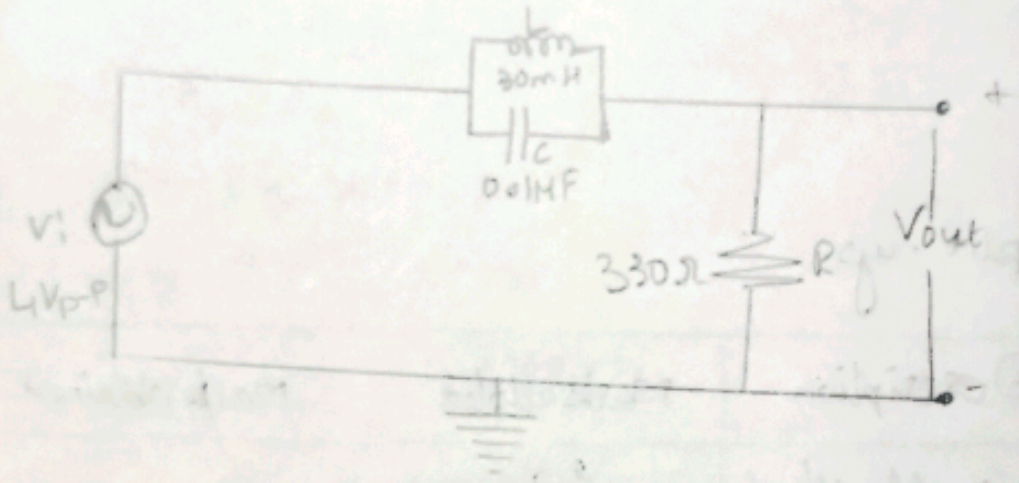
PROCEDURE:

1) Construct circuit shown in figure 1 on breadboard.
2) Connect CRO Ch1 to input and Ch2 to output.
3) Set the input voltage to 4Vp-p
4) Vary the frequency from 500Hz to 3kHz in small steps to get a maximum output voltage magnitude. The frequency at this maximum voltage V_{in} is the resonance frequency.
5) If $V_{cutoff} = \frac{V_{in}}{\sqrt{2}}$, vary the frequency again until the output voltage equals $V_{cut\,off}$. This frequency is the
6) Calculate the bandwidth by subtracting the 2 cutoff frequencies.
7) Calculate the Q factor.

Circuit Diagram:

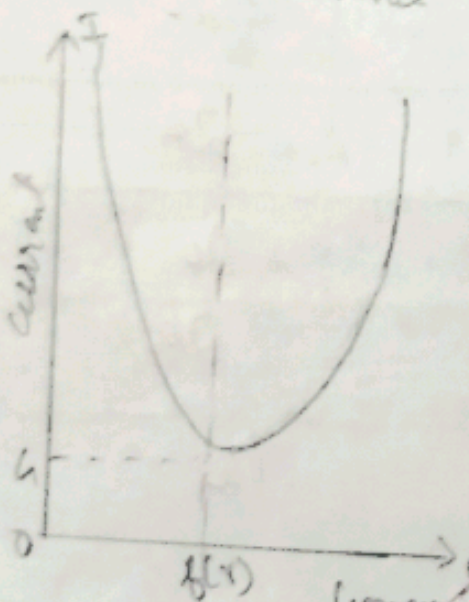


Series RLC Circuit

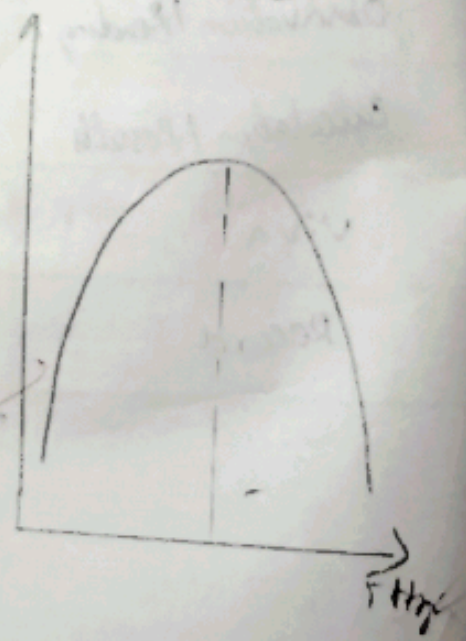


Parallel RLC Circuit

Parallel Resonance



Series resonance



TABULATION:

$V_{in} = 4\ V_{p-p}$

Series Resonance			Parallel Resonance		
Frequency (Hz)	Output Voltage (V)	output current (mA)	Frequency (Hz)	Output Voltage (V)	Output current (mA)
500	1.2	2	500	1.5	9
700	2.0	4	700	2.8	7
900	3.5	7	900	4.0	5
(Resonance) 1000	5.0	10	(Resonance) 1000	5.0	2
1100	3.6	7	1100	4.1	5

Calculation:-

$L = 100mH = 0.1H$, $C = 0.25\mu F = 0.25 \times 10^{-6} F$

$$f_0 = \frac{1}{2\pi\sqrt{(0.1)(0.25\times10^{-6})}} = \frac{1}{9.92\times10^{-4}} = 1000\ Hz$$

Maximum current (series) and maximum current (parallel) occur at 1000 Hz.

$$BW = \frac{R}{2\pi L} = \frac{330}{2\times3.14\times0.1} = \frac{330}{0.628} = 525.4$$

$$Q = \frac{f_0}{BW} = \frac{1000}{525.4} = 1.903$$

viii) Repeat step: 1-2 for circuit in fig 2.

FORMULA:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$BW = f_2 - f_1$$

$$Q = f_0/BW$$

Mark split-up.

S.NO	Description	Marks alloted	Marks obtained
1	Experimental setup	20	20
2	Observations	20	20
3	Calculation/Results	20	20
4	Viva	20	20
5	Record	20	20
	Total	100	100

Result:-

Thus the phenomenon of resonance in RLC circuits was studied of the following were determined using Multisim simulator.

Series RLC circuit:

Resonant frequency: 1000 Hz

Bandwidth: 525.4

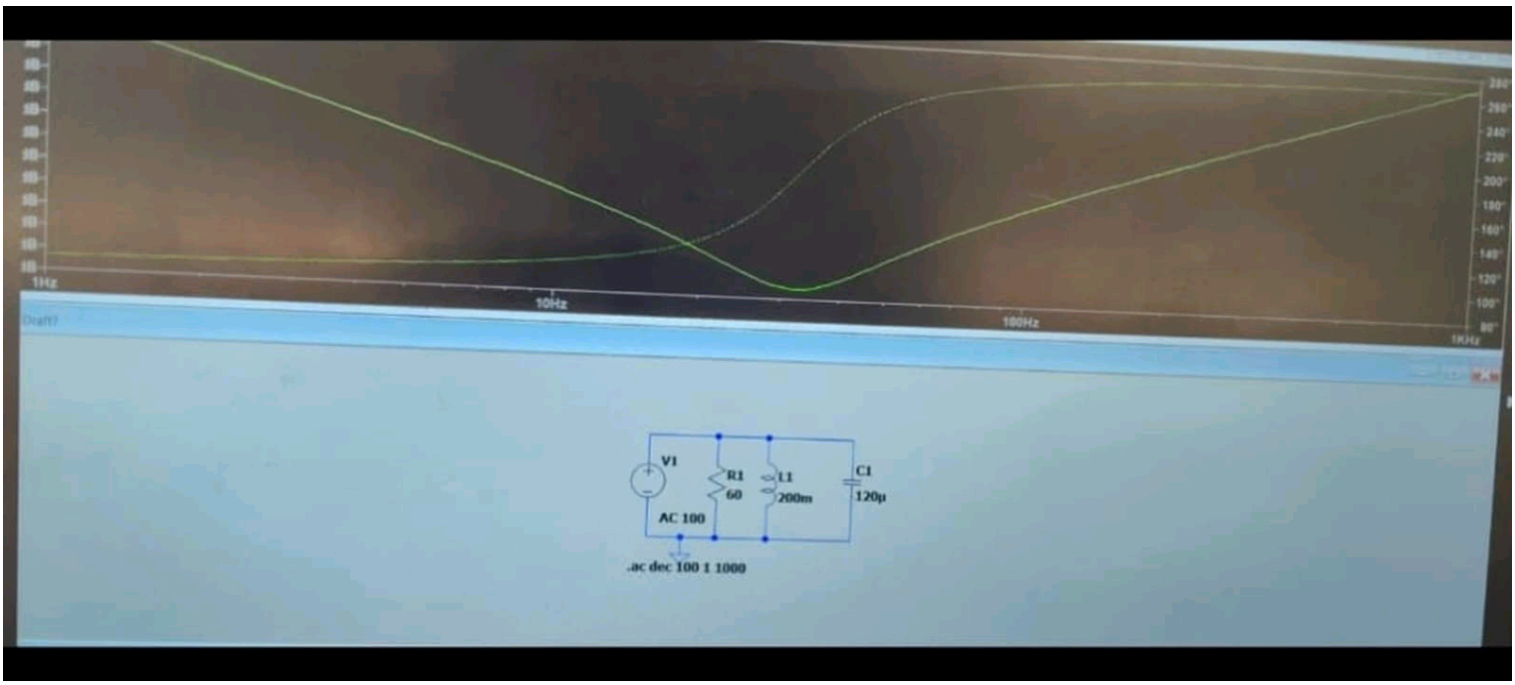
Q factor: 1.903

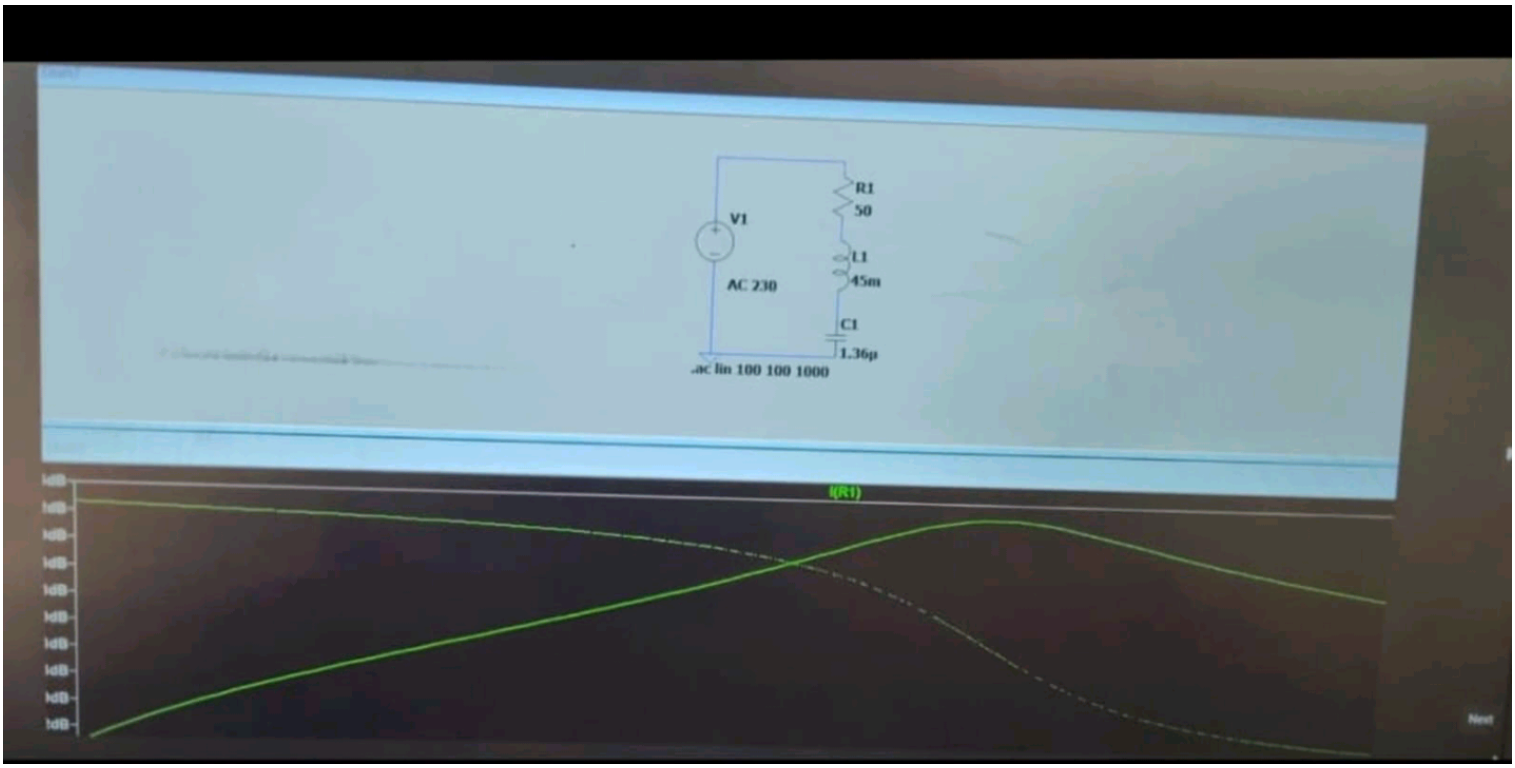
Parallel RLC circuit:

Resonant frequency: 1000 Hz

Band width: 525.4

Q factor = 1.903


1Hz
10Hz
100Hz
1KHz
V1
AC 100
R1
60
L1
200m
C1
120µ
.ac dec 100 1 1000


V1
AC 230
R1
50
L1
45m
C1
1.36µ
.ac lin 100 100 1000
I(R1)